

Maternal Haplogroup

You descend from a long line of female ancestors that can be traced back to eastern Africa over 150,000 years ago. These are the people of your maternal line, and your maternal haplogroup sheds light on their story.



Randy, your maternal haplogroup is H13a1a1a.

As our ancestors ventured out of eastern Africa, they branched off in diverse groups that crossed and recrossed the globe over tens of thousands of years. Some of their migrations can be traced through haplogroups, families of lineages that descend from a common ancestor. Your maternal haplogroup can reveal the path followed by the women of your maternal line.

Migrations of Your Maternal Line



180,000 Years Ago

Haplogroup L

If every person living today could trace his or her maternal line back over thousands of generations, all of our lines would meet at a single woman who lived in eastern Africa between 150,000 and 200,000 years ago. Though she was one of perhaps thousands of women alive at the time, only the diverse branches of her haplogroup have survived to today. The story of your maternal line begins with her.

65,000 Years Ago

Haplogroup L3

Your branch of L is haplogroup L3, which arose from a woman who likely lived in eastern Africa between 60,000 and 70,000 years ago. While many of her descendants remained in Africa, one small group ventured east across the Red Sea, likely across the narrow Bab-el-Mandeb into the tip of the Arabian Peninsula.

59,000 Years Ago

Haplogroup N

Your story continues with haplogroup N, one of two branches that arose from L3 in southwestern Asia. Researchers have long debated whether they arrived there via the Sinai Peninsula, or made the hop across the Red Sea at the Bab-el-Mandeb. Though their exact routes are disputed, there is no doubt that the women of haplogroup N migrated across all of Eurasia, giving rise to new branches from Portugal to Polynesia.

57,000 Years Ago

Haplogroup R

One of those branches is haplogroup R, which traces back to a woman who lived soon after the migration out of Africa. She likely lived in southwest Asia, perhaps in the Arabian peninsula, and her descendants lived and migrated alongside members of haplogroup N. Along the way, R gave rise to a number of branches that are major haplogroups in their own right.

18,000 Years Ago

Haplogroup H

While some members of R traveled far and wide, some remained in the Middle East for tens of thousands of years. Haplogroup H arose among the latter group, from a woman who likely lived less than 18,000 years ago. Her descendants expanded dramatically to the north after the Ice Age, and eventually reached from Arabia to the western fringes of Siberia.

10,000 Years Ago

Origin and Migrations of Haplogroup H13a1

Your maternal line stems from a branch of H called H13a1. Haplogroup H13a1 traces back to a common ancestor who lived nearly 10,000 years ago, likely in the Caucasus. Its parent branch is H13, a relatively rare lineage that hasn't expanded far beyond its origin in the Middle East. H13a1 is one of the exceptions to this pattern. People bearing H13a1 appear to have migrated westward along the Mediterranean coast of Europe near the beginning of the Neolithic, when the first farmers began to expand from the Caucasus to western Europe. Their migration has left a trace in the populations of mainland Italy, Sardinia and Iberia, where members of H13a1 are found at low levels.

H13a1a1a

4,500 Years Ago

Your maternal haplogroup, H13a1a1a, traces back to a woman who lived approximately 4,500 years ago.

That's nearly 180 generations ago! What happened between then and now? As researchers and citizen scientists discover more about your haplogroup, new details may be added to the story of your maternal line.

Today

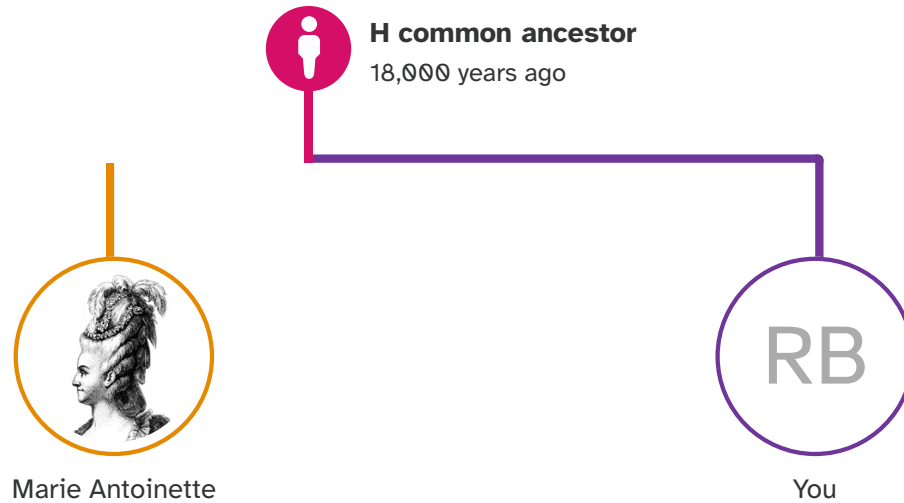
H13a1a1a is frequent among 23andMe customers.

Today, you share your haplogroup with all the maternal-line descendants of the common ancestor of H13a1a1a, including other 23andMe customers.

1 in 310

23andMe customers share your haplogroup assignment.

Haplogroup H and the Royal Lines of Europe



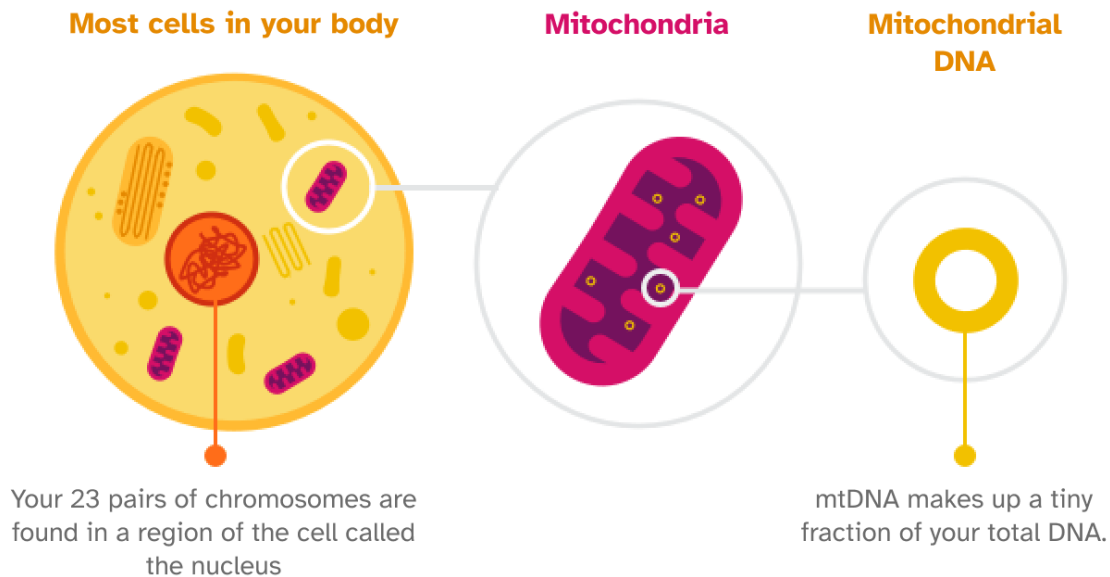
Because it is so dominant in the general European population, haplogroup H also appears quite frequently in the continent's royal houses. Marie Antoinette, an Austrian Hapsburg who married into the French royal family, inherited the haplogroup from her maternal ancestors. So did Prince Philip, Duke of Edinburgh, whose recorded genealogy traces his female line to Bavaria. Scientists also discovered that famed 16th century astronomer Nicolaus Copernicus traced his maternal lineages to haplogroup H.

The Genetics of Maternal Haplogroups

Mitochondrial DNA

Maternal haplogroups are determined by sets of genetic variants in a tiny, unusual loop of DNA called mitochondrial DNA (mtDNA). As the name suggests, mtDNA is found in the mitochondria, small but mighty structures inside our cells that turn fuel from the food we eat into energy.

Mitochondria evolved over billions of years from an independent bacterial cell that was engulfed by another cell. Instead of becoming lunch, the bacterium helped its new host use oxygen to produce energy. Over time it completely lost its independence and became an integrated part of the larger cell.

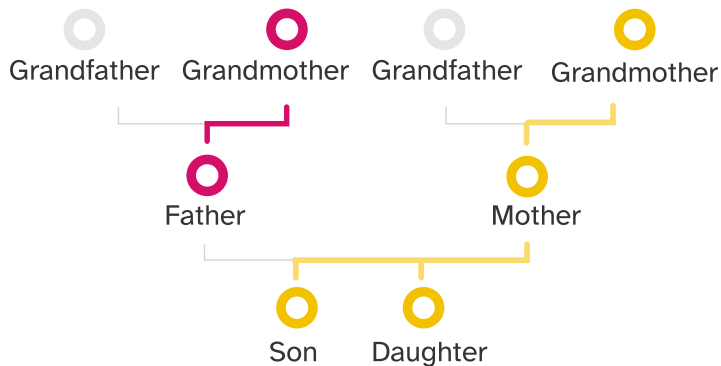


Maternal Haplogroup Tree

MtDNA is a powerful tool for tracing the history of maternal lines because of the way it is inherited: everyone has mtDNA, but only females pass it down to their children. So for example, an individual inherits a copy of their mother's mtDNA, who inherited it from their mother, who inherited it from theirs, and so on through the generations along an unbroken line of female ancestors.

The copies passed down are not always perfectly identical, however. Small typos in the mtDNA sequence occasionally occur, creating new genetic variants. Over many generations, these variants stack up in unique patterns that are carried by different maternal lines around the world.

Only mothers pass their mtDNA down to their children

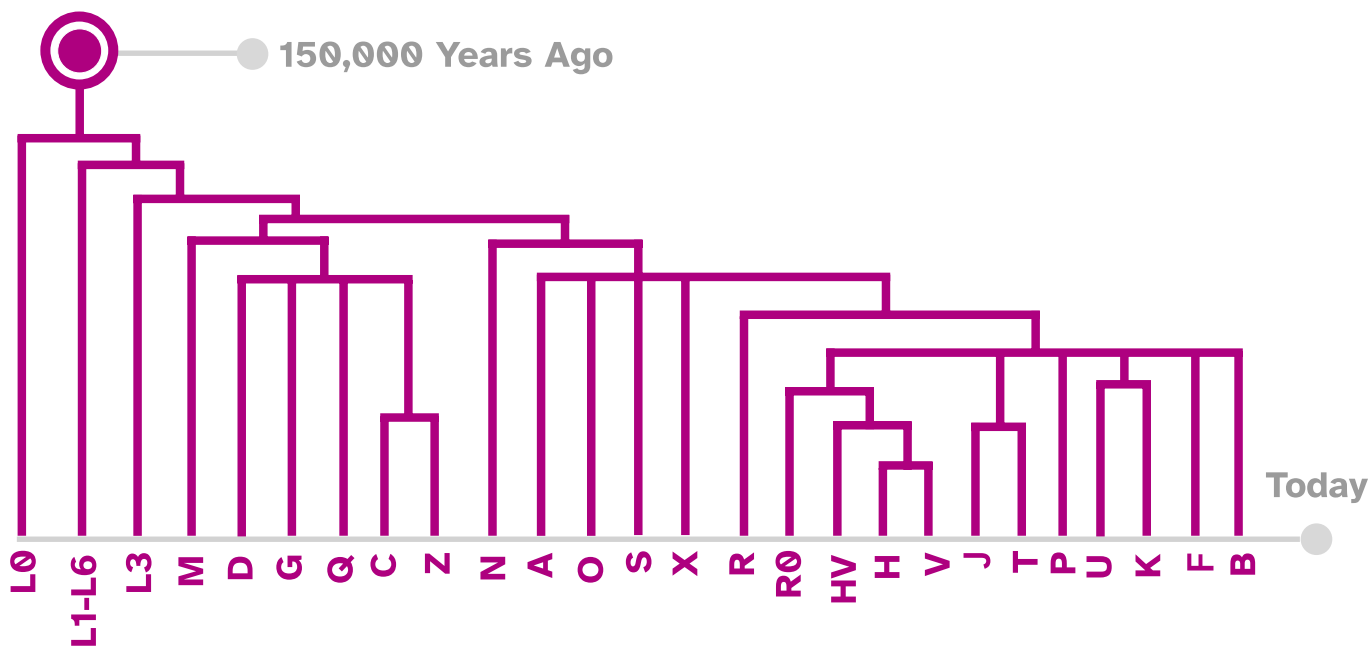


Maternal Haplogroup Tree

By comparing the mtDNA patterns from around the world, researchers identify families of maternal lines. All the lines within each family trace back to a single common ancestor, and share a set of mtDNA variants that they inherited from her.

In fact, when we look very far back in time, all the maternal lines around the world trace back to one woman! Along with her ancestors, she forms the root of a great tree that shows how all maternal lines are related. Each sub-family in this tree is called a "haplogroup" and named with a sequence of letters and numbers that reflect its location in the tree.

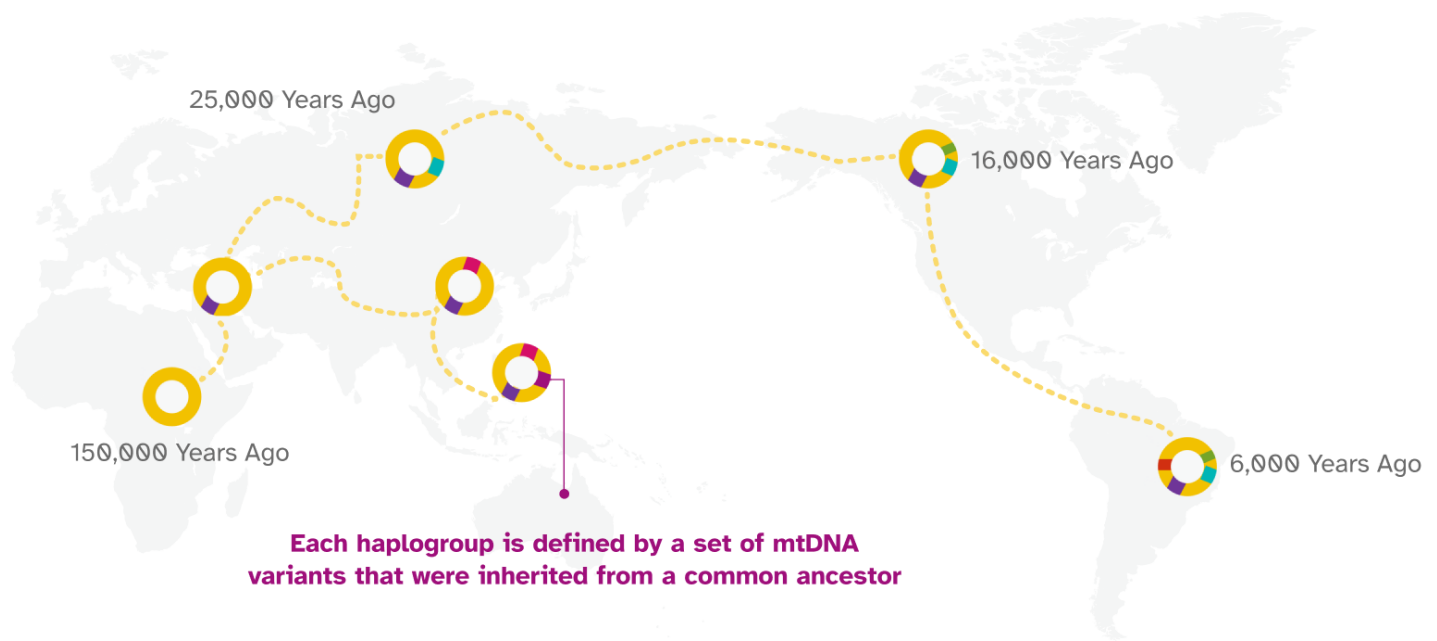
See your line in the tree of all maternal haplogroups.



Tracing Female Migrations

Next, geneticists study the relationships between haplogroups and compare them with the distribution of each group around the world. Because closely related haplogroups tend to share geographic roots, researchers can play a sophisticated version of connect-the-dots to estimate the origins and migration patterns of particular haplogroups.

Finally, combining this genetic evidence with data from other fields of study helps researchers place the story of each maternal line within the broader context of human history.



Do more with your Haplogroup results.

- Contribute to research and help us understand patterns of genetic variation around the world.
- Visit DNA Relatives to identify relatives that may be on your maternal line.

Scientific Details

Your haplogroup is determined by your mitochondrial DNA.

Each generation, females pass down copies of their mitochondrial DNA (mtDNA) to their children. While most of your genome exists in 23 pairs of chromosomes that exchange pieces between generations in a process called recombination, mtDNA is transmitted unshuffled. Because of this unusual pattern of inheritance, mtDNA contains rich information about maternal lineages.

A small number of DNA changes, called mutations, generally occur from one generation to the next. Because mtDNA does not recombine between generations, these mutations accumulate in patterns that uniquely mark individual lineages. Scientists can compare the sequence differences that result by constructing a tree. This tree shows how maternal lineages relate to one another, including the observation that they all share a most recent common ancestor approximately 150,000 to 200,000 years ago.

The term "haplogroup" refers to a family of lineages that share a common ancestor and, therefore, a particular set of mutations. We identify your haplogroup by determining which branches of the mtDNA tree correspond to your DNA. Because more closely related lineages tend to share geographic roots, your haplogroup can provide insight into the origins of some of your ancient maternal-line ancestors.

Maternal haplogroups are named with sequences of letters and numbers that reflect the structure of the tree and how the branches relate to one another.

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Change Log

Your report may occasionally be updated based on new information. This Change Log describes updates and revisions to this report.

Date	Change
May 8, 2017	The standalone Maternal Haplogroup report was created, featuring new design elements and content.
Oct. 21, 2015	Haplogroups report created.

Randy Buchanan's Report, printed on 2025-02-18 UTC



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